PAINT EDITOR · BACKDROPS

Create your own maze as a Scratch backdrop

📜 SCRIPT REFERENCE

Draw the Maze

STEPS TO FOLLOW

1. Open Scratch → delete cat
2. Click stage backdrop thumbnail → "Paint" (pencil icon)
3. Rectangle tool with solid dark blue fill
4. Draw thick rectangles for walls
5. Leave 30–40px corridors
6. Draw outer border first → back-arrow to return

💡 TIP

Keep paths wide! If they are too narrow, the hero will get stuck in the walls.

🤔 DID YOU KNOW?

In Scratch, "touching colour" checks the exact pixel colour — that's why solid, consistent wall colours are so important.

SPRITES · SIZE

Add a small hero sprite that fits inside the maze corridors

STEPS TO FOLLOW

1. Choose a Sprite → pick small character (Avery or arrow)
2. Set Size to 25–30
3. Drag to maze start
4. Note the x and y coordinates for next step

💡 TIP

After placing the hero, check that it's smaller than the maze corridors. If it looks cramped, reduce the size further.

🤔 DID YOU KNOW?

You can click anywhere on the stage while editing to see those x/y coordinates in the bottom-right of the editor.

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EVENTS · MOTION · Y-AXIS

Press the up and down arrow keys to move the hero vertically through the corridor

SCRATCH BLOCKS — ON THE HERO SPRITE

```
when 🏁 clicked
go to x: (-190) y: (150)
forever
  if <key [up arrow] pressed?> then
    change y by (4)
  if <key [down arrow] pressed?> then
    change y by (-4)
```

💡 TIP

"change y by (4)" moves UP because positive y is upward in Scratch. Negative y moves down — just like a maths graph!

🤔 DID YOU KNOW?

In Scratch, y=0 is the centre of the stage. y=180 is the very top; y=−180 is the very bottom.

2D VECTORS · 4-WAY INPUT

Add left and right arrow keys to complete full 4-direction movement

📜 SCRIPT REFERENCE

Move All 4 Directions

SCRATCH BLOCKS — ON THE HERO SPRITE

```
when 🏁 clicked
go to x: (-190) y: (150)
forever
  if <key [up arrow] pressed?> then
    change y by (4)
  if <key [down arrow] pressed?> then
    change y by (-4)
  if <key [left arrow] pressed?> then
    change x by (-4)
  if <key [right arrow] pressed?> then
    change x by (4)
```

💡 TIP

Use the same speed value (4) for all four directions so movement feels balanced.

🤔 DID YOU KNOW?

Vectors power every game engine, physics simulator, animation system, and robotics controller in existence.

COLOUR SENSING · PIXEL DETECTION

Detect when the hero touches a wall — say "Ouch!" as proof it works

SCRATCH BLOCKS — ADD INSIDE THE FOREVER LOOP

```
(at the end of the forever loop)
  if <touching color [■ dark blue]?> then
    say [Ouch! 🧱] for (0.5) secs
```

💡 TIP

The eyedropper only works when you click it first, THEN click the wall on the stage. The colour must match exactly.

🤔 DID YOU KNOW?

"touching color?" reads actual pixel data on screen — making the maze artwork itself the collision system. No coordinate math needed!

ROLLBACK PATTERN · COLOUR SENSING

Reverse each move if the hero touches a wall — the "rollback" pattern

SCRATCH BLOCKS — REPLACE EACH DIRECTION BLOCK

```
if <key [up arrow] pressed?> then
  change y by (4)
  if <touching color [■ dark blue]?> then
    change y by (-4) ← rollback!
... repeat for all 4 directions
```

💡 TIP

Do the EXACT same pattern for all 4 directions — up/down use y, left/right use x. Each direction gets its own immediate rollback.

🤔 DID YOU KNOW?

Database transactions, Git commits, undo/redo stacks, and save files all use rollback — "save state, try action, restore if bad."

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MULTIPLE SPRITES · PARALLEL SCRIPTS

Place a spinning star at the maze exit — it runs its own code independently

📜 SCRIPT REFERENCE

Add the Exit Star

SCRATCH BLOCKS — ON THE STAR SPRITE

– STAR: when 🏁 clicked →

```
forever
  turn (4) degrees
```

💡 TIP

Drag the star to the very end of your maze path — players must navigate all the way through to reach it.

🤔 DID YOU KNOW?

Multiple "when 🏁 clicked" scripts run simultaneously. This is called parallel execution — the same concept used in async programming, game engines, and operating systems.

WIN CONDITIONS · BOOLEAN LOGIC

Celebrate when the hero reaches the exit star — stop the game on victory

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — ON THE STAR SPRITE (SECOND SCRIPT)

– STAR: when 🏁 clicked (win script) →

```
forever
  if <touching [Hero]?> then
    say [You escaped! 🎉] for (3) seconds
    stop [all]
```

💡 TIP

The star now has TWO scripts — one spinning forever, one watching for the hero. Both run at the same time!

🤔 DID YOU KNOW?

Achievement systems, form validation, access control — all evaluate a boolean condition, then branch to a reward or error path.

VARIABLES · COUNTDOWN TIMER

Add a 30-second countdown; game over if time runs out

SCRATCH BLOCKS — TIMER SCRIPT (ANY SPRITE)

when 🏁 clicked (timer script)

```
set [Timer] to (30)
repeat (30)
  wait (1) seconds
  change [Timer] by (-1)
say [Time's up! ⏰ ] for (2) seconds
stop [all]
```

💡 TIP

To make it harder, reduce the timer to 20 or 15 seconds. To make it easier, increase it to 60.

🤔 DID YOU KNOW?

Login session timeouts, OTP codes, rate limiters, and cooking timers all work exactly like this countdown pattern.